Project Proposal

Campus Spaces Occupancy Predictor

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Campus Spaces Occupancy Predictor

The idea I had for the project is a campus space occupancy webapp that helps students estimate how crowded study areas, libraries, dining locations, and parking lots are before they waste time going themselves. The application would combine live user reports with historical usage patterns to estimate current occupancy and give predictions for how busy a location is likely to be soon. The main problem this project solves is that students often have little information about how crowded a campus location will be until they arrive. This can lead to wasted time walking between buildings, difficulty finding study space, or long waits at dining locations. My project would focus on giving students a simple way to make better decisions about where to go.

Major features:

- A campus map showing supported locations and estimated occupancy
- Simple user-submitted crowd reports such as low, moderate, or high
- Historical occupancy trends by location, day, and time
- Occupancy predictions based on previous patterns and recent reports

The application would use TypeScript with React for the web interface and a backend built with Node.js. I would use PostgreSQL for storing locations, reports, and historical occupancy data, along with a mapping library such as Leaflet and pandas for occupancy prediction. The intended users are students who regularly use campus study areas, dining facilities, and common spaces. I believe the project is genuinely useful and is feasible within a one semester timeframe.